Ravi Sekhar Kommuri

Data Engineer

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Summary

- Results-driven and detail-oriented **Data Engineer with 4+ years of experience** in architecting, building, and optimizing scalable data pipelines and analytics solutions in agile enterprise environments.
- Proficient in developing robust **ETL/ELT processes** using tools like **Apache Spark, Kafka, Airflow, dbt**, and cloud services including **AWS and Azure**.
- Strong expertise in **Python, SQL, and R,** with a solid foundation in data modeling, data warehousing, and real-time streaming. Adept at working with relational and **NoSQL databases such as MySQL, PostgreSQL, Oracle**, and **MongoDB,** ensuring high availability and data integrity.
- Skilled in data visualization and storytelling using **Tableau, Power BI, and Excel**, translating complex datasets into actionable insights for business stakeholders.
- Hands-on experience with **big data platforms** (Hadoop, Beam), **containerization (Docker)**, and **CI/CD pipelines** using **GitHub, Jenkins**, and **Terraform**. Well-versed in **SDLC**, **Agile**, and **Waterfall** methodologies, with strong communication, documentation, and cross-functional collaboration skills.
- Committed to data privacy and compliance with frameworks like **HIPAA**, **PCI DSS**, and **RBAC**, incorporating **data masking**, **tokenization**, and **data lineage** practices. Known for delivering high-performance, production-ready solutions that enhance business intelligence and decision-making across domains.

Skills

Methodologies:	SDLC, Agile, Waterfall
Programming Languages:	Python, SQL, R, Scala
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn
Visualization Tools:	Tableau, Power BI, Excel (Pivot Tables, VLOOKUP)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook
Cloud Platforms:	AWS, Azure
Database:	MySQL, Oracle SQL, PL/SQL, PostgreSQL, MongoDB
Data Engineering Concepts:	Apache Spark, Hadoop, Apache Kafka, Apache Iceberg, Apache Beam, dbt, ETL/ELT, PySpark
Generative AI:	GPT-4, BERT, LLaMA, LangGraph, RAG, Prompt Engineering, Tokenization
Other Technical Skills:	DAX, SAS, JIRA, SAP, SSIS, SSRS, Machine Learning Algorithms, Probability distributions, Hypothesis Testing, Regression Analysis, Advanced Analytics, Data Mining, Data Visualization, Data warehousing, Data transformation, Data Storytelling, Clustering, Classification, Regression, A/B Testing, Forecasting & Modelling, Data Cleaning, Data Wrangling, Ad Hoc Analysis, Project Management, Data Presentation, Requirement Gathering, Root Cause Analysis, Quantitative Analytics, Docker, Big Data & AI Integration, PCI DSS, Data Lineage & Masking, RBAC, CI/CD Pipeline
Version Control Tools:	Git, GitHub
Operating Systems:	Windows, Linux, Mac OS

Experience

Data Engineer | HCA Healthcare, TX — **Aug 2024 – Present**

- Designed and implemented a scalable **data lakehouse architecture** for HCA Healthcare, ingesting over 2 TB of structured and semi-structured clinical, claims, provider, and operational data across 180+ hospitals and 2,300 care sites nationwide.
- Built modular, version-controlled **data pipelines using Airflow and dbt**, orchestrating hourly incremental loads from 15+ upstream systems including Epic/Cerner EHRs, claims adjudication platforms, HL7/FHIR feeds, and pharmacy managers (~25M rows/day).
- Created parameterized **SQL queries** and stored procedures to support ad-hoc analysis, cohort selection, patient segmentation, and clinical KPI computation, improving query performance and analysis turnaround time by **18%**.

- Optimized **ETL performance** by **32%** using **Apache Spark on AWS EMR**, applying partition pruning, z-ordering, and **S3** I/O optimizations to accelerate processing of high-volume EHR and claims workloads.
- Developed and deployed machine learning models for **patient readmission prediction, sepsis early-warning detection, LOS forecasting, and risk stratification**, using cleaned and feature-engineered datasets from the scalable data lakehouse.
- Developed interactive **Power BI dashboards** for hospital leadership to monitor KPIs such as patient readmission rates, length of stay (LOS), care variability, sepsis alerts, and provider performance reducing reporting SLAs from 48 hours to under 3 hours.
- Strengthened data governance by implementing **data quality checks** (PHI validation, drift detection, referential integrity) and enforcing **PHI masking/anonymization** policies within transformation layers.
- Collaborated with **Clinical Analytics, and Actuarial teams** to deliver analytics-ready datasets supporting risk stratification, care pathway optimization, claims audit, and value-based care programs enabling $3.4M/year in operational savings and cost avoidance.

Data Engineer | Cognizant, India **Jul 2020 – Aug 2023**

- Architected and deployed a high-throughput **ETL pipeline** using **Apache Kafka and Azure Event Hubs**, processing of financial and market data daily with **99.9% reliability**, enabling real-time analytics for trading, compliance, and risk.
- Built scalable transformation pipelines using **PySpark in Azure Databricks**, integrating complex data sources including trades, portfolios, counterparty risk, and market data, supporting advanced modeling and regulatory reporting.
- Designed detailed **ER diagrams and star schema models** for financial instruments, trades, client relationships, and asset classes, facilitating faster reporting and clearer data lineage for audit trails.
- Increased **prediction accuracy by 20%** by training machine learning models on **Azure Machine Learning** with **scikit-learn**, predicting credit risk, portfolio drawdown, and trading anomalies across millions of records.
- Created interactive **Tableau dashboards** to visualize P&L performance, VaR exposure, counterparty credit limits, and trade reconciliation metrics, improving business stakeholder engagement by **50%** and cutting report delivery time by **15%**.
- Enabled **schema-less ingestion of unstructured financial documents** (regulatory filings, trader notes, compliance flags) into **MongoDB**, enriching data lake assets for exploratory analysis.
- Used **MapReduce** to perform batch tokenization and de-tokenization of sensitive client data in archived records, improving data privacy compliance by **15%** during historical data audits and reducing manual intervention in audit trails by **40%.**
- Developed **predictive models** for financial operations such as **capital adequacy forecasts and intraday liquidity usage**, achieving **18% forecast accuracy** in high-volatility scenarios.
- Integrated **trade, CRM, and compliance systems into a centralized Azure Synapse data warehouse**, enabling cross-functional analytics across front office, middle office, and risk teams.
- Orchestrated **100+ ETL workflows using Azure Data Factory and Apache Airflow on AKS**, ensuring timely data movement and transformation with **high availability and automated recovery**.
- Implemented **hybrid data architecture** using on-prem **Hadoop and Azure Blob Storage** with **PolyBase and data pipelines**, enabling seamless movement of large datasets between cloud and on-prem environments.

Education

Master of Science in Information Technology & Management - Campbellsville University, Louisville, Kentucky, USA
Bachelor of Technology in Computer Science and Engineering - Gudlavalleru Engineering College, JNTUK, India

Certifications

- Certified for “Python for Data Science”, a course issued by Smart Bridge Powered by IBM Developer Skills Network.
- Certified for “Machine Learning with Python”, a course issued by Smart Bridge Powered by IBM Developer Skills Network.